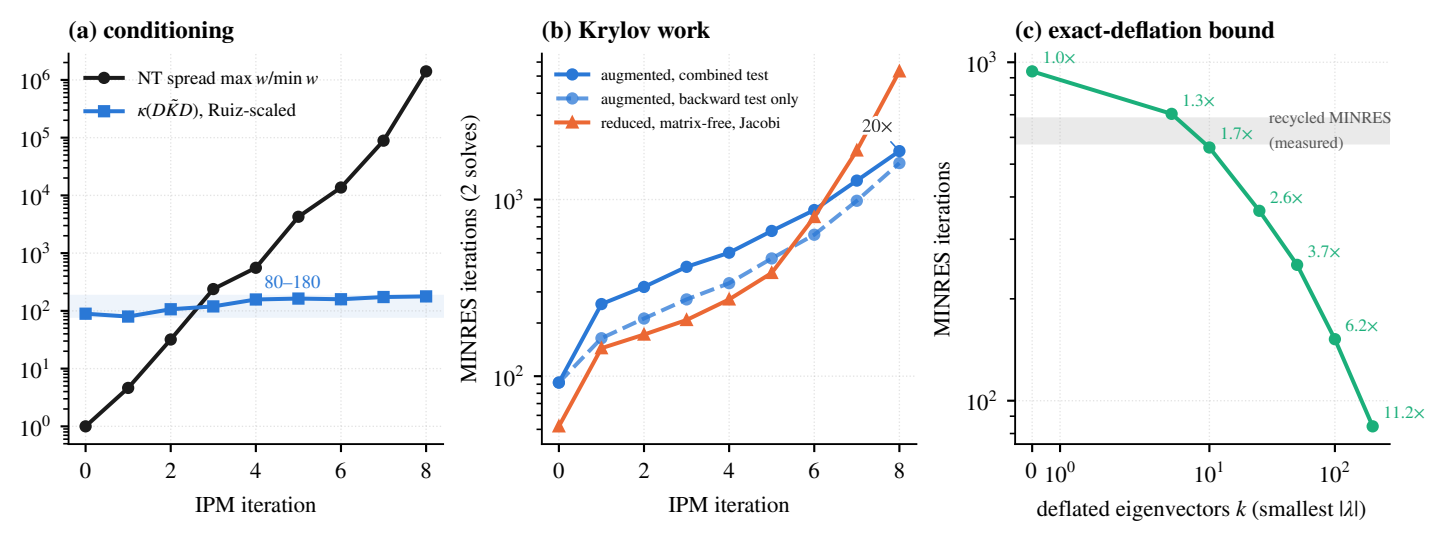
(a) conditioning
NT spread max w/min w
κ(DKD), Ruiz-scaled
80–180
IPM iteration
(b) Krylov work
augmented, combined test
augmented, backward test only
reduced, matrix-free, Jacobi
20×
MINRES iterations (2 solves)
IPM iteration
(c) exact-deflation bound
1.0×
1.3×
1.7×
2.6×
3.7×
6.2×
11.2×
recycled MINRES
(measured)
MINRES iterations
deflated eigenvectors k (smallest |λ|)